MA 104

Lec 8

Review of Power Series

- A power series is an infinite series of the form

$$\sum_{m=0}^{\infty} a_m (x-x_0)^m = a_0 + a_1(x-x_0) + a_2(x-x_0)^2 + \cdots\cdots.$$

Here the consts $a_0, a_1 \ldots$ are called the coefficients of the series, & the const. x_0 is called the center of the series.

- A power series $\sum_{m=0}^{\infty} a_m (x-x_0)^m$ is said to converge at a point x if the sequence $\{S_N(x)\}$ of partial sums

$$S_N(x) = \sum_{m=0}^{N} a_m (x-x_0)^m$$

is convergent, i.e., $\lim_{N\to\infty} S_N(x)$ exist.

✓ Notice the series certainly converges for $x = x_0$.

- The series $\sum_{m=0}^{\infty} a_m (x-x_0)^m$ is said to converge absolutely at a point x if the series $\sum_{m=0}^{\infty} |a_m (x-x_0)^m| = \sum_{m=0}^{\infty} |a_m| |x-x_0|^m$ converges.

Note. If a series converges absolutely, then the series also converges.

- If a power series $\sum_{n=0}^{\infty} a_n (x-x_0)^n$ converges at $x=x_1$, it converges absolutely for $|x-x_0| < |x_1-x_0|$, & if diverges at $x=x_1$, it diverges for $|x-x_0| > |x_1-x_0|$.

- $\exists$ a positive integer ρ, called the radius of convergence, s.t. $\sum_{m=0}^{\infty} a_m (x-x_0)^m$ converges for $|x-x_0| < \rho$ and diverges for $|x-x_0| > \rho$.
The interval $|x-x_0| < \rho$ is called the interval of converges.

- The root test states that the series converges if

$$c := \limsup_{n\to\infty} \sqrt[n]{|a_n(x-x_0)^n|} = \limsup_{n\to\infty} \left(\sqrt[n]{|a_n|}\right)|x-x_0| < 1$$

& diverges if $c > 1$. So $\rho = \dfrac{1}{\limsup_{n\to\infty} \sqrt[n]{|a_n|}}$.

- The ratio test states that the series converges if

$$L := \lim_{n\to\infty} \frac{|a_{n+1}(x-x_0)^{n+1}|}{|a_n(x-x_0)^n|} < 1$$

& diverges if $L > 1$. So $\rho = \lim_{n\to\infty} \left|\dfrac{a_n}{a_{n+1}}\right|$.

Rmk. Suppose $\sum_{n=0}^{\infty} a_n(x-x_0)^n$ & $\sum_{n=0}^{\infty} b_n(x-x_0)^n$ converges to $f(x)$ & $g(x)$ for $|x-x_0| < \rho$ for $\rho > 0$. Then the series

$$f(x) \pm g(x) = \sum_{n=0}^{\infty} (a_n \pm b_n)(x-x_0)^n \quad \& \quad f(x)g(x) = \sum_{n=1}^{\infty} c_n (x-x_0)^n,$$

where $c_n = a_0 b_n + a_1 b_{n-1} + \cdots + a_n b_0$ converges at least for $|x-x_0| < \rho$.

□ Suppose $\sum_{n=0}^{\infty} a_n(x-x_0)^n$ converge to $f(x)$ for $|x-x_0|<\rho$, $\rho>0$. Then the fun f is contn & has derivatives of all orders for $|x-x_0|<\rho$. Moreover, f', f'', .. can be computed by differentiating the series term wise, i.e.,

$$f'(x) = a_1 + 2a_2(x-x_0) + \cdots + na_n(x-x_0)^{n-1} + \cdots$$
$$= \sum_{n=1}^{\infty} na_n(x-x_0)^{n-1}$$

$$f''(x) = 2a_2 + 6a_3(x-x_0) + \cdots + n(n-1)a_n(x-x_0)^{n-2} + \cdots$$
$$= \sum_{n=2}^{\infty} n(n-1)a_n(x-x_0)^{n-2}.$$

$\vdots$

& each series converges absolutely for $|x-x_0|<\rho$.

- The value of a_n is given by : $a_n = \dfrac{f^{(n)}(x_0)}{\lfloor n}$.

The series $f(x) = \sum_{n=0}^{\infty} a_n (x-x_0)^n = \sum_{n=0}^{\infty} \dfrac{f^{(n)}(x_0)}{\lfloor n} (x-x_0)^n$ is called the <u>Taylor series</u> for the fu^n $f(x)$ about $x = x_0$.

- A fu^n $h(x)$ is called <u>analytic</u> at a point $x = x_0$ if it can be represented by a power series (Taylor series) in powers of $x - x_0$ with a positive radius of convergence.

<u>Eg.</u> e^x, polynomials, $\cos x$, $\sin x$, f/g (f & g are poly^{n}s) are analytic fu^n except at the zeros of $g(x)$.

- <u>Vanishing of all coefficients:</u> If $\rho > 0$ & $\sum_{n=0}^{\infty} a_n (x-x_0)^n = 0$ $\forall\ |x - x_0| < \rho$, then $a_n = 0$ $\forall\ n \geq 0$.

Shifting of index of summation:

It is immaterial which letter is used for the index of the summation. Sometime it is convenient to make changes of summation indices in calculating series solu^n of diff. equ^n.

Eg.
$$\sum_{n=2}^{\infty} a_n x^n = \sum_{m=0}^{\infty} a_{m+2} x^{m+2} = \sum_{n=0}^{\infty} a_{n+2} x^{n+2}$$

put $n-2=m$
$\Rightarrow n = m+2$

put $m=n$

1st term corresponds to $n=2$

1st term corresponds to $n=0$.

But, both contains precisely the same terms.

Eg. 1. Consider the series $\sum_{n=1}^{\infty} (-1)^{n+1} n (x-2)^n$. $\rightarrow a_n = (-1)^{n+1} n$

Notice $$\rho = \lim_{n\to\infty} \frac{|a_n|}{|a_{n+1}|} = \mathop{Lt}_{n\to\infty} \frac{n}{n+1} = \mathop{Lt}_{n\to\infty} \left(\frac{1}{1+1/n}\right) = 1$$

So the series converges (absolutely) if $|x-2|<1$ or $1<x<3$ & diverges for $|x-2|>1$.

Eg. 2 Consider the series $\sum_{n=0}^{\infty} \frac{(-1)^n}{k^n} x^{2n}$. $\rightarrow a_n = \frac{(-1)^{n/2}}{k^{n/2}}$, $x_0 = 0$

Notice $$\rho = 1 \Big/ \limsup_{n\to\infty} |a_n|^{1/n} = \overset{\displaystyle\lim_{n\to\infty}}{\overset{\shortparallel}{\limsup_{n\to\infty}}} \left(\frac{1}{|k|^{1/2}}\right)^{-1} = \sqrt{|k|}.$$

So the series converges if $|x| < \sqrt{|k|}$ & diverges if $|x| > \sqrt{|k|}$.

Convergence Radius $R = \infty, 1, 0$

For all three series let $m \to \infty$

$$e^x = \sum_{m=0}^{\infty} \frac{x^m}{m!} = 1 + x + \frac{x^2}{2!} + \cdots, \qquad \left|\frac{a_{m+1}}{a_m}\right| = \frac{1/(m+1)!}{1/m!} = \frac{1}{m+1} \to 0, \qquad R = \infty$$

$$\frac{1}{1-x} = \sum_{m=0}^{\infty} x^m = 1 + x + x^2 + \cdots, \qquad \left|\frac{a_{m+1}}{a_m}\right| = \frac{1}{1} = 1, \qquad R = 1$$

$$\sum_{m=0}^{\infty} m!x^m = 1 + x + 2x^2 + \cdots, \qquad \left|\frac{a_{m+1}}{a_m}\right| = \frac{(m+1)!}{m!} = m + 1 \to \infty, \qquad R = 0.$$

Convergence for all x ($R = \infty$) is the best possible case, convergence in some finite interval the usual, and convergence only at the center ($R = 0$) is useless.

Q. Find the Taylor series for $f(z) = \frac{e^z}{1-z}$ around $z=0$.

$\rightarrow$ Note $f(z) = g(z)\,h(z)$, where

$$g(z) = e^z = 1 + z + z^2/2! + z^3/3! + \cdots \quad \forall z \in \mathbb{R}$$

$$h(z) = \frac{1}{1-z} = 1 + z + z^2 + z^3 + \cdots \quad \forall |z| < 1.$$

Thus $f(z) = 1 + (1+1)z + (1+1+1/2!)z^2 + \left(1+1+\frac{1}{2!}+\frac{1}{3!}\right)z^3 + \cdots$

$\forall |z| < 1 = \rho$ (Ex.)

Eg. The series $\sum_{n=0}^{\infty} x^n/n!$ converges at $x=-1$, but does not converge absolutely. Notice $1 - 1/2 + 1/3 - 1/4 \cdots = \ln 2$, but $1 + 1/2 + 1/3 + \cdots$ diverges.

Question asked by a student (Check!)

Suppose $\sum_{n=0}^{\infty} a_n x^n$ converges to $f(x)$ for $|x|<\rho$.

Then $f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$. If the series $\sum_{n=1}^{\infty} n a_n x^{n-1}$ converges absolutely at some x, can we say $\sum_{n=0}^{\infty} a_n x^n$ converges absolutely?

$$\Rightarrow |a_1||x| + |a_2||x^2| + |a_3||x^3| + \cdots$$

$$= |x|\left(|a_1| + |a_2||x| + |a_3||x^2| + \cdots\right)$$

$$\leq |x|\underbrace{\left(|a_1| + 2|a_2||x| + 3|a_3||x^2| + \cdots\right)}_{\text{converges (given)}}$$

Since $g(x) = x = 0 + 1.x + 0.x^2 + \cdots$ also converges absolutely $\forall x \in \mathbb{R}$

(comparison test) $\Rightarrow$ the series $|a_1||x| + |a_2||x^2| + \cdots$ converges to L (say) $\forall |x|<\rho$

$\Rightarrow |a_0| + |a_1||x| + |a_2||x^2| + \cdots$ converges to $|a_0| + L$. $\forall |x|<\rho$

$\Rightarrow$ The series $\sum a_n x^n$ converges absolutely.

Consider the ODE: $y'' + p(x)y' + q(x) = r(x)$. — (1)

Q. When do a power series solu^n exist?

Ans. If p, q, r are analytic fu^{n}s.

Th^m: (Existence of Power Series Solu^{n}s).

If p, q, r in (1) are analytic at $x = x_0$, then every solu^n of (1) is analytic at $x = x_0$ & can thus be represented by a power series in powers of $x - x_0$ with radius of convergence $\rho > 0$.

Eg: Consider the ODE: $y'' + y = 0$ $\quad -\infty < x < \infty$. (A)

$\rightarrow p(x) = 0$ & $q(x) = 1$

We look for a solu^n of the form of a power series $y(x) = \sum_{n=0}^{\infty} a_n x^n$ — (1) about $x_0 = 0$ with the radius of converges $\rho > 0$.

Differentiating (1) we get: $y' = \sum_{n=1}^{\infty} n a_n x^{n-1}$ — (2)

$$y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} \quad — (3)$$

Substituting (1), (2) & (3) in (A) we get

$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} + \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\Rightarrow \sum_{n=0}^{\infty} \left[(n+2)(n+1) a_{n+2} + a_n \right] x^n = 0$$

recurrence relation

$$\Rightarrow (n+2)(n+1) a_{n+2} + a_n = 0 \quad \forall\ n = 0, 1, 2, \ldots$$

By induction, $a_{2k} = \dfrac{(-1)^k}{(2k)!} a_0$ & $a_{2k+1} = \dfrac{(-1)^k}{(2k+1)!} a_1$.

Thus $y(x) = \sum_{n=0}^{\infty} a_n x^n = \sum_{k=0}^{\infty} a_{2k} x^{2k} + \sum_{k=0}^{\infty} a_{2k+1} x^{2k+1}$

$$= \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} a_0 x^{2k} + \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} a_1 x^{2k+1}$$

$$= a_0 \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} x^{2k} + a_1 \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1}$$

$$= a_0 \cos x + a_1 \sin x$$

Ex. Use ratio test to show that each series is convergent $\forall x$.